JEE–MAIN EXAMINATION – JANUARY 2026

(HELD ON WEDNESDAY 28th JANUARY 2026) **TIME : 9:00 AM TO 12:00 NOON**

MATHEMATICS — TEST PAPER WITH ANSWER

SECTION-A

1. If $g(x) = 3x^2 + 2x - 3, f(0) = -3$ and $4g(f(x)) = 3x^2 - 32x + 72$, then $f(g(2))$ is equal to:

(1) $\frac{25}{6}$ (2) $-\frac{25}{6}$

(3) $\frac{7}{2}$ (4) $-\frac{7}{2}$

Ans. (3)

2. The value of $\sum_{k=1}^{\infty}(-1)^{k+1}\left(\frac{k(k+1)}{k!}\right)$ is :

(1) 2/e (2) 1/e

(3) $\sqrt{e}$ (4) e/2

Ans. (2)

3. Let $y = x$ be the equation of a chord of the circle C_1 (in the closed half-plane $x \geq 0$) of diameter 10 passing through the origin. Let C_2 be another circle described on the given chord as its diameter. If the equation of the chord of the circle C_2, which passes through the point (2, 3) and is farthest from the center of C_2, is $x + ay + b = 0$, then $a - b$ is equal to :

(1) 10 (2) –6

(3) –2 (4) 6

Ans. (3)

4. If $\frac{\tan(A-B)}{\tan A} + \frac{\sin^2 C}{\sin^2 A} = 1$, $A, B, C \in \left(0, \frac{\pi}{2}\right)$, then

(1) tan A, tan C, tan B are in G.P.

(2) tan A, tan B, tan C are in G.P.

(3) tan A, tan C, tan B are in A.P.

(4) tan A, tan B, tan C are in A.P.

Ans. (1)

5. Let z be a complex number such that $|z - 6| = 5$ and $|z + 2 - 6i| = 5$.

Then the value of $z^3 + 3z^2 - 15z + 141$ is equal to

(1) 42 (2) 37

(3) 50 (4) 61

Ans. (3)

6. Let ABC be an equilateral triangle with orthocenter at the origin and the side BC on the line $x + 2\sqrt{2}y = 4$. If the co-ordinates of the vertex A are (α, β), then the greatest integer less than or equal to $|\alpha + \sqrt{2}\beta|$ is

(1) 2 (2) 3

(3) 5 (4) 4

Ans. (4)

7. Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Let x be the number of 9-digit numbers formed using the digits of the set S such that only one digit is repeated and it is repeated exactly twice. Let y be the number of 9-digit numbers formed using the digits of the set S such that only two digits are repeated and each of these is repeated exactly twice. Then,

(1) 29x = 5y (2) 45x = 7y

(3) 21x = 4y (4) 56x = 9y

Ans. (3)

8. Let $S = \{x^3 + ax^2 + bx + c : a, b, c \in N \text{ and } a, b, c \leq 20\}$ be a set of polynomials. Then the number of polynomials in S, which are divisible by $x^2 + 2$, is

(1) 20 (2) 6

(3) 120 (4) 10

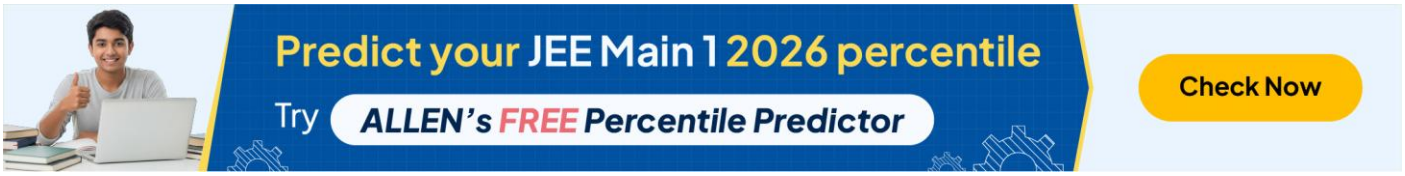
Ans. (4)

9. A bag contains 10 balls out of which k are red and (10 – k) are black, where $0 \leq k \leq 10$. If three balls are drawn at random without replacement and all of them are found to be black, then the probability that the bag contains l red and 9 black balls is :

(1) $\frac{7}{11}$ (2) $\frac{7}{55}$

(3) $\frac{7}{110}$ (4) $\frac{14}{55}$

Ans. (4)



10. If α, β, where $\alpha < \beta$, are the roots of the equation $\lambda x^2 - (\lambda + 3)x + 3 = 0$ such that $\frac{1}{\alpha} - \frac{1}{\beta} = \frac{1}{3}$, then the sum of all possible values of λ is :

(1) 6 (2) 2

(3) 4 (4) 8

Ans. (1)

11. If $\int\left(\frac{1-5\cos^2 x}{\sin^5 x \cos^2 x}\right)dx = f(x) + C$ where C is the constant of integration, then $f\left(\frac{\pi}{6}\right) - f\left(\frac{\pi}{4}\right)$ is equal to

(1) $\frac{1}{\sqrt{3}}\left(26+\sqrt{3}\right)$ (2) $\frac{4}{\sqrt{3}}\left(8-\sqrt{6}\right)$

(3) $\frac{1}{\sqrt{3}}\left(26-\sqrt{3}\right)$ (4) $\frac{2}{\sqrt{3}}\left(4+\sqrt{6}\right)$

Ans. (2)

12. Let f be a polynomial function such that $f(x^2 + 1) = x^4 + 5x^2 + 2$, for all $x \in \mathbb{R}$.

Then $\int_0^3 f(x)dx$ is equal to

(1) $\frac{41}{3}$ (2) $\frac{33}{2}$

(3) $\frac{27}{2}$ (4) $\frac{5}{3}$

Ans. (2)

13. The area of the region $R=\{(x, y): xy \le 8, 1 \le y \le x^2, x \ge 0\}$ is

(1) $\frac{1}{3}\left(49\log_e(2)-15\right)$

(2) $\frac{2}{3}\left(20\log_e(2)+9\right)$

(3) $\frac{2}{3}\left(24\log_e(2)-7\right)$

(4) $\frac{1}{3}\left(40\log_e(2)+27\right)$

Ans. (3)

14. The value of

$$\lim_{x\to 0}\frac{\log_e\left(\sec(ex).\sec\left(e^2x\right)\cdot\ldots\cdot\sec\left(e^{10}x\right)\right)}{e^2 - e^{2\cos x}}$$

is equal to

(1) $\frac{(e^{10}-1)}{2e^2(e^2-1)}$ (2) $\frac{(e^{20}-1)}{2e^2(e^2-1)}$

(3) $\frac{(e^{20}-1)}{2(e^2-1)}$ (4) $\frac{(e^{10}-1)}{2(e^2-1)}$

Ans. (3)

15. The mean and variance of 10 observations are 9 and 34.2, respectively. If 8 of these observations are 2, 3, 5, 10, 11, 13, 15, 21, then the mean deviation about the median of all the 10 observations is

(1) 5 (2) 4

(3) 6 (4) 7

Ans. (1)

16. Let A, B and C be three 2×2 matrices with real entries such that $B = (I + A)^{-1}$ and $A + C = I$. If $BC = \begin{bmatrix} 1 & -5 \\ -1 & 2 \end{bmatrix}$ and $CB\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 12 \\ -6 \end{bmatrix}$, then $x_1 + x_2$ is

(1) 2 (2) 0

(3) –2 (4) 4

Ans. (2)

17. The common difference of the A.P.: $a_1, a_2, \ldots, a_m$ is 13 more than the common difference of the A.P.: $b_1, b_2, \ldots, b_n$. If $b_{31} = -277$, $b_{43} = -385$ and $a_{78} = 327$, then a_1 is equal to

(1) 21 (2) 24

(3) 19 (4) 16

Ans. (3)



18. If the distances of the point (1, 2, a) from the line $\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{1}$ along the lines

$L_1:\frac{x-1}{3}=\frac{y-2}{4}=\frac{z-a}{b}$ and

$L_2:\frac{x-1}{1}=\frac{y-2}{4}=\frac{z-a}{c}$ are equal,

then a + b + c is equal to

(1) 7 (2) 5

(3) 6 (4) 4

Ans. (1)

19. For three unit vectors $\vec{a},\vec{b},\vec{c}$ satisfying $|\vec{a}-\vec{b}|^2+|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2=9$ and

$|2\vec{a}+k\vec{b}+k\vec{c}|=3,$

the positive value of k is :

(1) 3 (2) 6

(3) 4 (4) 5

Ans. (4)

20. Let y = y(x) be the solution of the differential equation

$x\frac{dy}{dx}-\sin 2y=x^3(2-x^3)\cos^2 y,\ x\neq 0.$

If y (2) = x, then tan(y(1)) is equal to

(1) $\frac{3}{4}$ (2) $\frac{7}{4}$

(3) $-\frac{7}{4}$ (4) $-\frac{3}{4}$

Ans. (2)

SECTION-B

21. In a G.P., if the product of the first three terms is 27 and the set of all possible values for the sum of its first three terms is $\mathbb{R}-(a, b)$, then a^2+b^2 is equal to ____.

Ans. (90)

22. For some $\theta\in\left(0,\frac{\pi}{2}\right)$, let the eccentricity and the length of the latus rectum of the hyperbola $x^2-y^2\sec^2\theta=8$ be e_1 and ℓ_1, respectively, and let the eccentricity and the length of the latus rectum of the ellipse $x^2\sec^2\theta+y^2=6$ be e_2 and ℓ_2, respectively. If $e_1^2=e_2^2\left(\sec^2\theta+1\right)$, then $\left(\frac{\ell_1\ell_2}{e_1e_2}\right)\tan^2\theta$ is equal to _____.

Ans. (8)

23. If $k=\tan\left(\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\left(\frac{2}{3}\right)\right)+\tan\left(\frac{1}{2}\sin^{-1}\left(\frac{2}{3}\right)\right)$ then the number of solutions of the equation $\sin^{-1}(kx-1)=\sin^{-1}x-\cos^{-1}x$ is _____.

Ans. (1)

24. The value of $\sum_{r=1}^{20}\left(\left|\sqrt{\pi\left(\int_0^r x\,|\sin\pi x|\,dx\right)}\right|\right)$ is ____.

Ans. (210)

25. Let PQR be a triangle such that $\overrightarrow{PQ}=-2\hat{i}-\hat{j}+2\hat{k}$ and $\overrightarrow{PR}=a\hat{i}+b\hat{j}-4\hat{k}$, a, b $\in\mathbb{Z}$. Let S be the point on QR, which is equidistant from the lines PQ and PR. If $|\overrightarrow{PR}|=9$ and $\overrightarrow{PS}=\hat{i}-7\hat{j}+2\hat{k}$, then the value of 3a – 4b is ___.

Allen Ans. (Bonus)

NTA Ans. (37)



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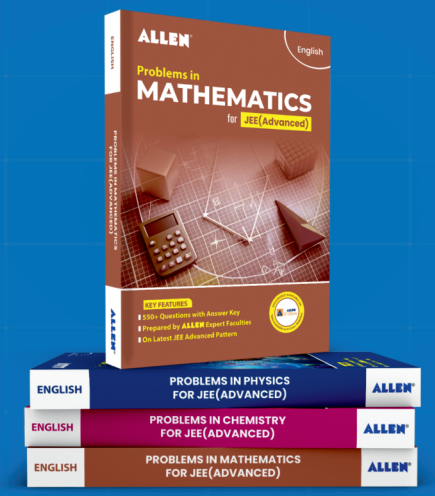


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